

BUILDING THE STANDARD MODEL LAGRANGIAN FROM THE HQIV DISCRETE ACTION: A SECTOR-BY-SECTOR LEAN BRIDGE

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ABSTRACT. The Standard Model Lagrangian density,

$$\mathcal{L}_{\text{SM}} = -\frac{1}{4}(B_{\mu\nu}^2 + W_{\mu\nu}^2 + G_{\mu\nu}^2) + i\bar{\psi}\gamma^\mu D_\mu\psi + (D_\mu\Phi)^\dagger(D^\mu\Phi) - V(\Phi) - (y_f \bar{f}\Phi f + \text{h.c.}),$$

is the page-long expression that defines high-energy particle physics. In this manuscript we package, in machine-checked Lean (zero `sorry`), a sector-by-sector identification of every term of $\mathcal{L}_{\text{SM}}$ with projections of the HQIV *discrete* action

$$\mathcal{S}_{\text{HQIV}} = \text{action.O.Maxwell.general } J_{\text{src}} A \phi_{\text{val}} + S_{\text{HQVM grav}} \phi \rho_m \rho_r$$

already proved in `Hqiv/Physics/Action.lean` and its companions. The identification is constructive: the gauge kinetic block is the sectorial projection of the abelian octonion kinetic density L_O^{kin} through the force-sector map `0_component_to_sector`; the fermion kinetic plus minimal coupling is the abelian source coupling L_O^{sc} for the triality-summed SM current; the Higgs kinetic and potential are read off the octonion-scalar norm `scalarNormSq` and the `higgsPotential` of `WeakHiggsFrom0MaxwellScaffold` evaluated at the lock-in vev `lockinVev`; the Yukawa couplings are the two-step resonance ladder `resonance_k_tau_mu`, `resonance_k_mu_e` of `ChargedLeptonResonance` renormalized against the Planck-unit τ mass. The headline theorem `SM_Lagrangian_from_HQIV_discrete_action` (proved in `Hqiv/Physics/StandardModelLagrangianFromDiscreteAction.lean`) asserts the five sector equalities simultaneously. The parameter witness `SM_Lagrangian_parameter_witness` certifies that only three lattice rationals ($\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$) and the two resonance steps enter; no PDG / $\overline{\text{MS}}$ current-quark masses, no fitted Yukawa table, no external lattice inputs are used. The paper follows the audit discipline of the companion variational manuscript [1]: every claim cites an exact Lean identifier; what is *not* proved (anomaly cancellation, measure choice, the explicit non-abelian holonomy upgrade) is named explicitly. A continuum extension of the SM Lagrangian is *not* an open problem in this framework: it is a calculation approximation only, and at the load-bearing steps (the curvature imprint $\delta_E(m) \propto 1/(m+1)$, the resonance ladder $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor $m = 4$, and the rationals $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$) a literal continuum version would erase the discrete index doing the predictive work. The single principled exception is the $\mathfrak{so}(8)$ closure $G_2 \cup \{\Delta\}$, a de-facto continuum algebra obtained from discrete inputs.

Patch QFT and discrete numerics (reader contract). HQIV is formulated as a *patch theory*: quantum fields, actions, and physical readouts are attached to discrete patches—shell indices $m \in \mathbb{N}$, finite spacetime index types (e.g. `Fin 4`), and horizon-limited charts—rather than to a hypothetical fundamental smooth spacetime obtained as a mesh-refinement or ultraviolet limit. Accordingly, when this text refers to “QFT,” it means *patch QFT*: patching rules, finite measurement ledgers, and variational identities on the discrete spine; it is not the claim that the theory is *defined* by recovering ordinary continuum quantum field theory through such a limit. The numbers that admit the sharpest statements—mass and coupling ladders, curvature ratios, shell thermodynamics, and rational witnesses in `HQIV_LEAN`—are objects of *discrete mathematics* on $\mathbb{N}$ (combinatorial growth,

cumulative simplex ledgers) together with the ordered field $\mathbb{R}$ as used in the proof assistant for analysis, bounds, and phenomenological display. Smooth-manifold or continuum field notation, when it appears, is a *translation layer* for comparison with standard physics literature; it does not supersede the discrete definitions.

A continuum is not required, and in places would destroy these results. The discrete patch layer is *not* a numerical approximation to a more fundamental continuum theory awaiting future construction; it is the ontology. Where the literature treats “the continuum limit” as the goal, HQIV treats it as a *calculation approximation* for comparison with standard formulas. Statements that read in earlier drafts as “the continuum extension is an open problem” are to be reread as “the continuum form is an optional convenience for comparison; the proved discrete statement is what carries the physics.” In several load-bearing places—the curvature imprint $\delta_E(m)$ at shell m , the harmonic ladder masses $m_\tau \rightarrow m_\mu \rightarrow m_e$, the lock-in vev $v = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$, the proton anchor at $m = \text{referenceM} = 4$, the baryon-asymmetry $\eta \propto 6^7 \sqrt{3}/(m+1)$, and the rational coefficients $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ —taking a literal $m \rightarrow \infty$ or sub-Planck continuum limit would *erase* the discrete index that is doing the predictive work. Continuum forms of these objects are therefore not preferred refinements; they are degenerate, information-losing readouts of the same patch data.

The one exception: $\mathfrak{so}(8)$ as a de-facto continuum algebra from a discrete construction. There is a single, principled exception to the discrete-only stance: the closed Lie algebra $\mathfrak{so}(8) \supseteq G_2 \cup \{\Delta\}$ generated from the Fano octonion left-multiplication table and the phase-lift generator $\Delta \in (e_1, e_7)$. Although $\mathfrak{so}(8)$ is a continuous (28-dimensional real) Lie algebra, it is built here entirely from *discrete* ingredients (the seven imaginary octonion units, a finite Fano table, and one $\pi/2$ rotation), and its closure to 28 dimensions is a finite linear-algebra certificate (see the `Hqiv.S08Closure / Hqiv.S08ClosureInterface` stack). The Lie-group structure is therefore a *de-facto continuum algebra obtained from a discrete construction*: continuous in parameter space, but with no continuum spacetime, no continuum measure, and no UV limit attached. When this manuscript or its companions write $\exp(t\Delta) \in \text{SO}(8)$ or treat $\mathfrak{so}(8)$ rotations as smooth, those are statements internal to a finite-dimensional Lie group whose generators are certified by discrete data; they are *not* a continuum-spacetime commitment and do not require a continuum-QFT scaffolding.

1. SCOPE AND CLAIM DISCIPLINE

1.1. Predecessor: discrete O–Maxwell action. [1] packages the discrete octonion-indexed gauge potential A^a_μ (channels $a \in \text{Fin } 8$, spacetime indices $\mu, \nu \in \text{Fin } 4$), field strength $F^a_{\mu\nu}(A) = A^a_\nu - A^a_\mu (\text{F_from_A})$, the kinetic density $L_O^{\text{kin}}(A)$ (`L_0_kinetic`), the source interaction $L_O^{\text{src}}(J_{\text{src}}, A)$ (`L_0_source_general`), and the ϕ -coupling slot $L_O^\phi(A, \phi_{\text{val}})$ (`L_0_phi_coupling`) into the single Lagrangian density (`L_0_Maxwell_general`)

$$\mathcal{L}_O[J_{\text{src}}](A, \phi_{\text{val}}) = L_O^{\text{kin}}(A) + 4\pi L_O^{\text{src}}(J_{\text{src}}, A) + L_O^\phi(A, \phi_{\text{val}}).$$

The gravitational constraint scalar $S_{\text{HQVM}_{\text{grav}}}(\phi, \rho_m, \rho_r) = (3 - \gamma_{\text{HQIV}})\phi^2 - 8\pi G_{\text{eff}}(\phi)(\rho_m + \rho_r)$ (`S_HQVM_grav`) vanishes iff the HQVM Friedmann equation holds (`S_HQVM_grav_zero_iff_Friedmann`). The total cell `action_total_general` is $\mathcal{L}_O + S_{\text{HQVM}_{\text{grav}}}$. [2] adds the charged-lepton resonance ladder `resonance_k_tau_mu`, `resonance_k_mu_e`, the τ anchor `m_tau_P1`, and the generation-level mass functional `smMassFromGeometryLabel` that ties an SM flavour label to a Planck-unit rest mass with no PDG input. [3] and the `Hqiv/Algebra/SMEembedding.lean` module fix the $\text{SU}(2)_L$

and hypercharge generators inside the certified $G_2 \cup \{\Delta\} \Rightarrow \mathfrak{so}(8)$ closure, exhaust the standard branching rules on 8_s , and certify the three triality irreps that we read as generation labels.

1.2. What this paper adds. The new Lean module is `Hqiv/Physics/StandardModelLagrangianFromDiscreteAction.lean` (zero sorry, builds clean under `lake build`); its principal contributions are:

- **Sector splitting of the abelian octonion kinetic** (Sect. 2): the lemma `L_0_kinetic_eq_sum_of_sector_pieces` writes $L_O^{\text{kin}} = L_{O,\text{EM}}^{\text{kin}} + L_{O,\text{Weak}}^{\text{kin}} + L_{O,\text{Strong}}^{\text{kin}}$ under the explicit force-sector assignment `O_component_to_sector` from `Hqiv/Physics/Forces.lean`; sector aggregates `L_SM_B_kinetic`, `L_SM_W_kinetic`, `L_SM_G_kinetic` project to the textbook `B`, `W`, `G` kinetic blocks; the SM gauge total equals L_O^{kin} exactly (`L_SM_gauge_kinetic_eq_L_0_kinetic`).
- **Triality-summed fermion coupling** (Sect. 3): per-generation fermion currents $J(\text{rep8V})$, $J(\text{rep8SPlus})$, on the 8_s carrier are summed by two applications of the proved linearity `L_0_source_general_add_J` (`L_SM_three_generations_eq_total_source`).
- **Higgs as the octonion scalar** (Sect. 4): the Higgs kinetic block is the per-direction sum of `scalarNormSq`; the symmetry-breaking potential $\lambda(|\Phi|^2 - v^2)^2$ is the existing `higgsPotential` with $\lambda \mapsto \lambda_{\text{eff}}$ (`lambda_eff` in `SM_GR_Unification.lean`) and $v \mapsto \text{lockinVev}$ (`lockinVev`), expanded to the textbook $\lambda|\Phi|^4 - 2\lambda v^2|\Phi|^2 + \lambda v^4$ form (`L_SM_Higgs_potential_expanded`).
- **Yukawa from the resonance ladder** (Sect. 5): $y_f := \sqrt{2}m_f/v$ with $m_f = \text{smMassFromGeometryLabel } f$; the identity $y_f v/\sqrt{2} = m_f$ is proved (`y_SM_times_v_over_sqrt2_eq_mass`), and the explicit witness $y_f v/\sqrt{2} = m_\tau/\text{resonanceProduct}(\text{smGenerationIndex } f)$ exhibits the entire Yukawa block as the two-step ladder (`SM_Lagrangian_yukawa_resonance_witness`).
- **Assembled SM Lagrangian** (Sect. 6): the record `SM_Lagrangian` and its constructor `SM_Lagrangian.fromHQIV` package the five sector pieces; the headline theorem `SM_Lagrangian_from_HQIV_discrete_action` certifies the five sector equalities simultaneously; the total-action equality `HQIV_total_action_eq_SM_gauge_fermion_plus_phi_coupling_plus_grav` exhibits the SM gauge + fermion + ϕ -coupling + Friedmann split of the HQIV total cell when $J_{\text{src}} = \sum_{\text{gen}} J(\text{gen})$.
- **Parameter witness** (Sect. 7): `SM_Lagrangian_parameter_witness` certifies the three lattice rationals $\alpha = 3/5$, $\gamma = 2/5$, $\alpha_{\text{GUT}} = 1/42$ (`alpha_eq_3_5`, `gamma_eq_2_5`, `alpha_GUT_eq_1_42`); these together with m_τ^{Pl} and the two resonance steps are the entire parameter input to $\mathcal{L}_{\text{SM}}$.

1.3. What is and isn't proved. This module is a *structural bridge*, not a re-derivation of textbook field equations:

- **Proved (Lean):** the five sector equalities of `SM_Lagrangian_from_HQIV_discrete_action`; the total-action split of `HQIV_total_action_eq_SM_gauge_fermion_plus_phi_coupling_plus_grav`; the parameter witness; linearity of the Yukawa sum `L_SM_Yukawa_sum_eq_sum`.
- **Black-box ingredients:** the HQIV pieces L_O^{kin} , L_O^{src} , L_O^ϕ , S_{HQVMgrav} , `higgsPotential`, `lockinVev`, `smMassFromGeometryLabel`, `lambda_eff`, `O_component_to_sector` are taken as given from the cited modules; all `sm.*.from_HQIV` theorems are tautological equalities showing that the symbolic SM density exposed here is literally the corresponding HQIV expression. The translation to smooth-manifold notation (where the variational derivative of the discrete action is matched against the textbook Dirac equation) is provided in `Hqiv/Physics/ContinuumOmaxwellClosure.lean` and the continuum embedding files; this is a

comparison layer, not a foundational requirement — the load-bearing predictions of [1] live on the discrete spine and a literal continuum limit would destroy the shell index m they depend on.

- **Symbolic kinetic terms:** the fermion and Higgs kinetic shadows are kept on the octonion carrier $\Phi : \text{Fin } 8 \rightarrow \mathbb{R}$; no manifold integration or BRST quantisation is performed.
- **Non-abelian holonomy:** the per-sector gauge kinetic uses the abelian field strength $F^a_{\mu\nu} = A^a_\nu - A^a_\mu$. The non-abelian upgrade lives in `Hqiv/Physics/DiscretePlaquetteHolonomy.lean` and the `weakF` scaffold of `WeakHiggsFrom0MaxwellScaffold.lean`; the latter already implements the explicit commutator slot $F^{a,\text{weak}}_{\mu\nu} = (W^a_\nu - W^a_\mu) + g_w \text{comm}^a_{\mu\nu}$ (`weakF`, `L_weak_kinetic`).
- **No fits, no PDG.** The Yukawa block is read directly off the two-step resonance ladder; the Higgs vev is the lock-in $\text{lockinVev} = \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$. The reference shell $m = \text{referenceM} = 4$ enters only through these ladders; the proton anchor of [2] is preserved exactly.

2. SECTOR PROJECTION OF THE GAUGE KINETIC BLOCK

The HQIV abelian octonion kinetic density (`L_0_kinetic` of [1], kinetic density definition),

$$L_O^{\text{kin}}(A) = -\frac{1}{4} \sum_{a \in \text{Fin } 8} \sum_{\mu, \nu \in \text{Fin } 4} \frac{(F^a_{\mu\nu}(A))^2}{2},$$

splits into three sector pieces. The force-sector assignment is the O -component map

$$\text{O_component_to_sector}(a) = \begin{cases} \text{EM}, & a = 0, \\ \text{Weak}, & a \in \{1, 2, 3\}, \\ \text{Strong}, & a \in \{4, 5, 6, 7\}, \end{cases}$$

fixed in `Hqiv/Physics/Forces.lean`; component 0 in the quaternionic subalgebra reduces to classic Maxwell (`O_component_zero_is_EM`), the three remaining quaternionic components carry the weak-like sector, and the four octonion-extension components carry the colour-like sector.

Definition 2.1 (Sector projections of L_O^{kin}). *For $s \in \{\text{EM}, \text{Weak}, \text{Strong}\}$ and a potential A ,*

$$L_{O,s}^{\text{kin}}(A) := -\frac{1}{4} \sum_{a \in \text{Fin } 8} \chi_s(a) \sum_{\mu, \nu \in \text{Fin } 4} \frac{(F^a_{\mu\nu}(A))^2}{2},$$

where $\chi_s(a) = 1$ if `O_component_to_sector`(a) = s and zero otherwise (`L_0_kinetic_sector`).
Aggregating:

$$\mathcal{L}_B(A) := L_{O,\text{EM}}^{\text{kin}}(A), \quad \mathcal{L}_W(A) := L_{O,\text{Weak}}^{\text{kin}}(A), \quad \mathcal{L}_G(A) := L_{O,\text{Strong}}^{\text{kin}}(A)$$

(`L_SM_B_kinetic`, `L_SM_W_kinetic`, `L_SM_G_kinetic`).

Theorem 2.2 (Three-sector splitting, Lean). *For every A ,*

$$L_O^{\text{kin}}(A) = L_{O,\text{EM}}^{\text{kin}}(A) + L_{O,\text{Weak}}^{\text{kin}}(A) + L_{O,\text{Strong}}^{\text{kin}}(A).$$

This is `L_0_kinetic_eq_sum_of_sector_pieces`.

Sketch (Lean tactic translation). For each octonion channel $a \in \text{Fin } 8$, exactly one of the three indicator functions $\chi_{\text{EM}}(a), \chi_{\text{Weak}}(a), \chi_{\text{Strong}}(a)$ is one and the other two vanish (case split on

`0_component_to_sector`). Summing the channel-wise identity $\sum_{\mu,\nu} \frac{F^2}{2} = \chi_{\text{EM}}(a) \sum_{\mu,\nu} \frac{F^2}{2} + \chi_{\text{Weak}}(a) \sum_{\mu,\nu} \frac{F^2}{2} + \chi_{\text{Strong}}(a) \sum_{\mu,\nu} \frac{F^2}{2}$ over a and applying $-\frac{1}{4} \cdot$ yields the claim. Lean discharges the finite case split with `cases + simp` and the global sum with `Finset.sum_add_distrib`. $\square$

Corollary 2.3 (SM gauge kinetic equals HQIV octonion kinetic). $\mathcal{L}_B(A) + \mathcal{L}_W(A) + \mathcal{L}_G(A) = L_O^{\text{kin}}(A)$ (`L_SM_gauge_kinetic_eq_L_O_kinetic`).

Reading. The textbook SM gauge kinetic block $\mathcal{L}_{\text{gauge}} = -\frac{1}{4}(B_{\mu\nu}B^{\mu\nu} + W_{\mu\nu}^I W^{I\mu\nu} + G_{\mu\nu}^a G^{a\mu\nu})$ is the sector-projected aggregate of the single HQIV object L_O^{kin} when `0_component_to_sector` is the identification fixed by the algebra (colour-preferred axis and hypercharge block of `SMEembedding.lean`). No new coupling, no new field strength: the gauge-kinetic “page” of $\mathcal{L}_{\text{SM}}$ is one Lean identifier with a sector partition.

3. FERMION SECTOR: MINIMAL COUPLING AND THREE GENERATIONS

The textbook block $i\bar{\psi}\gamma^\mu D_\mu\psi$ summed over one chiral SM generation is supported on the 8-dimensional left-handed 8_s carrier of `Hqiv/Algebra/SMEembedding.lean`, plus 8 right-handed components in 8_c (see the branching rule `branching_rules_8s`, $8_s \rightarrow (3, 2, +\frac{1}{6}) \oplus (\bar{3}, 1, -\frac{2}{3}) \oplus (\bar{3}, 1, +\frac{1}{3}) \oplus (1, 2, -\frac{1}{2}) \oplus (1, 1, +1) \oplus (1, 1, 0)$ of `SMEembedding.lean`). At the abelian level needed for the present bridge, the Dirac bilinear is encoded as a current $J : \text{Fin } 8 \rightarrow \text{Fin } 4 \rightarrow \mathbb{R}$ on the 8_s carrier, and the minimal coupling reads off the proved $J \cdot A$ slot L_O^{src} .

Definition 3.1 (Per-generation minimal coupling). *For a triality generation label $\text{gen} \in \text{So8RepIndex}$ and a generation current J_{gen} ,*

$$L_{\text{fermion}}^{\text{SM}}(\text{gen}, J_{\text{gen}}, A) := L_O^{\text{src}}(J_{\text{gen}}, A)$$

(`L_SM_fermion_minimal_coupling`). *The three SM generations correspond to the three $\text{Spin}(8)$ irreps $8_v, 8_{s+}, 8_{s-}$ counted by `three_generations_from_triality_reps` (`card So8RepIndex = 3`).*

Theorem 3.2 (Triality-summed fermion coupling, Lean). *For any triality-indexed family of currents $J : \text{So8RepIndex} \rightarrow (\text{Fin } 8 \rightarrow \text{Fin } 4 \rightarrow \mathbb{R})$,*

$$\sum_{\text{gen} \in \{8_v, 8_{s+}, 8_{s-}\}} L_{\text{fermion}}^{\text{SM}}(\text{gen}, J(\text{gen}), A) = L_O^{\text{src}}\left(\sum_{\text{gen}} J(\text{gen}), A\right).$$

This is `L_SM_three_generations_eq_total_source`; its proof is two applications of the proved J -linearity `L_O_source_general_add_J` (Lemma 3.1 of [1]).

Remark 3.3 ($8_v, 8_{s+}, 8_{s-}$ vs. generation labelling). *The repository’s mass module `Hqiv/Physics/SM_GR_Unification.lean` assigns generation $1 \rightarrow 8_v$, generation $2 \rightarrow 8_{s+}$, generation $3 \rightarrow 8_{s-}$ via `smGenerationIndex`; this is a representation-counting fact (`three_generations_from_triality_reps`), not a mass theorem. The mass hierarchy itself is read off the harmonic ladder of `Hqiv/Physics/ChargedLeptonResonance.lean` via the `resonanceProduct` (Section 5 below); see the explicit discussion in the “Harmonic ladder vs. triality” note of `SMEembedding.lean`.*

Reading. The textbook fermion sector $\mathcal{L}_{\text{fermion}} = \sum_{\text{gen}} i\bar{\psi}_{\text{gen}}\gamma^\mu D_\mu\psi_{\text{gen}}$ is, at the bookkeeping level relevant to the discrete action, the abelian source coupling for the sum of the three triality-indexed currents on the 8_s carrier. The covariant derivative D_μ encodes the same gauge connection A that supplies the sector kinetic block of Section 2; no additional gauge structure is needed.

4. HIGGS SECTOR: OCTONION SCALAR, VEV, AND QUARTIC

The HQIV scalar lives on the same $\text{Fin } 8 \rightarrow \mathbb{R}$ carrier as the gauge potential (`OctonionScalar` in `WeakHiggsFromOMaxwellScaffold.lean`); the Higgs kinetic shadow of the textbook $(D_\mu \Phi)^\dagger (D^\mu \Phi)$ is read off the diagonal `scalarNormSq`, summed over spacetime directions; the symmetry-breaking potential is the existing `higgsPotential` at the calibrated lock-in vev.

Definition 4.1 (Higgs sector projection). *For a per-direction scalar slot $\Phi_{\text{dir}} : \text{Fin } 4 \rightarrow (\text{Fin } 8 \rightarrow \mathbb{R})$,*

$$\mathcal{L}_{\text{Higgs}}^{\text{kin}}(\Phi_{\text{dir}}) := \sum_{\mu \in \text{Fin } 4} \text{scalarNormSq}(\Phi_{\text{dir}}(\mu))$$

(`L_SM_Higgs_kinetic`). *For a scalar configuration Φ ,*

$$\mathcal{L}_{\text{Higgs}}^{\text{pot}}(\Phi) := \text{higgsPotential}(\lambda_{\text{eff}}, \text{lockinVev}, \Phi) = \lambda_{\text{eff}}(|\Phi|^2 - \text{lockinVev}^2)^2$$

(`L_SM_Higgs_potential`), *with $\lambda_{\text{eff}} \in \mathbb{R}$ from `Hqiv/Physics/SM_GR_Unification.lean` and the lock-in vev from `Hqiv/Physics/WeakHiggsFromOMaxwellScaffold.lean`,*

$$\text{lockinVev} := \sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}.$$

Theorem 4.2 (Textbook expanded form, Lean).

$$\mathcal{L}_{\text{Higgs}}^{\text{pot}}(\Phi) = \lambda_{\text{eff}} |\Phi|^4 - 2\lambda_{\text{eff}} \text{lockinVev}^2 |\Phi|^2 + \lambda_{\text{eff}} \text{lockinVev}^4.$$

This is `L_SM_Higgs_potential_expanded` (proved by `unfold plus ring`).

Remark 4.3 (Lock-in calibration of the vev). *The calibration*

$$\text{lockinVev}^2 = \eta_{\text{paper}} \quad (\text{when } \Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1)$$

is `lockinVev_sq_eq_eta_paper` in `Hqiv/Physics/WeakHiggsFromOMaxwellScaffold.lean`; the curvature ratio $\Omega_k(m_{\text{lockin}}; m_{\text{lockin}}) = 1$ is `omega_k_lockin_calibration` from `Hqiv/Physics/BaryogenesisCore.lean`. The lock-in vev is therefore not a fitted constant: it is the square root of the proved baryogenesis η -anchor of the paper, packaged once and reused unchanged in the Higgs sector.

Reading. The textbook Higgs “page” $\mathcal{L}_H = (D_\mu \Phi)^\dagger (D^\mu \Phi) - \mu^2 |\Phi|^2 + \lambda |\Phi|^4$ becomes, in HQIV notation, the diagonal sum of `scalarNormSq` (kinetic) plus the textbook expanded form of the proved `higgsPotential` with $(\lambda, v^2) = (\lambda_{\text{eff}}, \eta_{\text{paper}})$. The constant $\lambda_{\text{eff}} v^4$ is a global vacuum offset that does not enter equations of motion.

5. YUKAWA SECTOR FROM THE RESONANCE LADDER

The Yukawa block of $\mathcal{L}_{\text{SM}}$ is $\mathcal{L}_Y = -\sum_f y_f \bar{f} \Phi f + \text{h.c.}$. In HQIV the Yukawa coupling for each SM flavour is the textbook combination $y_f = \sqrt{2} m_f / v$ with $v = \text{lockinVev}$ and m_f not a free parameter:

$$m_f = \text{smMassFromGeometryLabel } f = m_\tau^{\text{Pl}} / \text{resonanceProduct}(\text{smGenerationIndex } f),$$

where `resonanceProduct` is built from the two charged-lepton resonance steps $k_{\tau\mu} = \text{resonance_k_tau_mu}$, $k_{\mu e} = \text{resonance_k_mu_e}$ of `ChargedLeptonResonance.lean`: generations $\{8_v, 8_{s+}, 8_{s-}\}$ get product $k_{\tau\mu} k_{\mu e}$, $k_{\tau\mu}$, 1 respectively.

Definition 5.1 (SM Yukawa coupling). *For an SM flavour label $\ell \in \text{SMassLabel}$ (`SMassLabel` in `SM_GR_Unification.lean`),*

$$y_{\text{SM}}(\ell) := \frac{\sqrt{2} \text{smMassFromGeometryLabel}(\ell)}{\text{lockinVev}}$$

(`y_SM`). *For a symbolic Dirac bilinear $B : \text{SMassLabel} \rightarrow \mathbb{R}$,*

$$\mathcal{L}_{\text{Yukawa}}^{\text{SM}}(\ell, B) := -y_{\text{SM}}(\ell) B(\ell)$$

(`L_SM_Yukawa_flavour`); *the all-flavour aggregate is the list sum over `all_SM_flavours` = [electron, muon, tau, up, down, strange, charm, bottom, top, ν_e , ν_μ , ν_τ] (12 entries, certified by `all_SM_flavours_length`).*

Theorem 5.2 (Yukawa identity, Lean). *For every SM flavour ℓ and every $v = \text{lockinVev} \neq 0$,*

$$\frac{y_{\text{SM}}(\ell) v}{\sqrt{2}} = \text{smMassFromGeometryLabel}(\ell) = m_\tau^{\text{Pl}} / \text{resonanceProduct}(\text{smGenerationIndex } \ell).$$

This is the pair (`y_SM_times_v_over_sqrt2_eq_mass`, `SM_Lagrangian_yukawa_resonance_witness`).

Theorem 5.3 (Yukawa linearity over flavours, Lean). *For any flavour list L and bilinear assignment B ,*

$$\sum_{\ell \in L} \mathcal{L}_{\text{Yukawa}}^{\text{SM}}(\ell, B(\ell)) = - \sum_{\ell \in L} y_{\text{SM}}(\ell) B(\ell).$$

This is `L_SM_Yukawa_sum_eq_sum`; its proof is a straight list induction that unfolds `L_SM_Yukawa_flavour` and closes with `ring`.

Remark 5.4 (No fitted Yukawa matrix). *The textbook SM stores 3×3 Yukawa matrices for u -, d -, ℓ -, and (with right-handed neutrinos) ν -type fermions, plus a 9-dimensional CKM/PMNS mixing block — in total $\mathcal{O}(20)$ real numbers fitted from PDG. In HQIV these are all replaced by the single rule $y_f = \sqrt{2} m_f / v$ with m_f from the resonance ladder. The mixing structure is the triality-orbit branching of `SMEembedding.lean`; mass hierarchies are the harmonic ladder, not a tuned Yukawa table.*

Reading. The textbook Yukawa “page” $-y_e \bar{L} \Phi e_R - y_u \bar{Q} \tilde{\Phi} u_R - y_d \bar{Q} \Phi d_R + \text{h.c.}$ (three generations) becomes, in HQIV notation, a list-sum over twelve flavours each weighted by $\sqrt{2} m_\tau^{\text{Pl}} / (\text{lockinVev} \cdot \text{resonanceProduct})$. Three numbers (m_τ^{Pl} , $k_{\tau\mu}$, $k_{\mu e}$) fix the entire Yukawa sector.

6. ASSEMBLED SM LAGRANGIAN AND HEADLINE THEOREMS

Definition 6.1 (`SM_Lagrangian` record (Lean)). *The structure `SM_Lagrangian` carries five real-valued fields ($\mathcal{L}_{\text{gauge}}$, $\mathcal{L}_{\text{fermion}}$, $\mathcal{L}_{\text{Higgs}}^{\text{kin}}$, $\mathcal{L}_{\text{Higgs}}^{\text{pot}}$, $\mathcal{L}_Y$). The total is*

$$\mathcal{L}_{\text{SM}}^{\text{total}} := \mathcal{L}_{\text{gauge}} + \mathcal{L}_{\text{fermion}} + \mathcal{L}_{\text{Higgs}}^{\text{kin}} - \mathcal{L}_{\text{Higgs}}^{\text{pot}} + \mathcal{L}_Y$$

(`SM_Lagrangian.total`).

Definition 6.2 (`SM_Lagrangian.fromHQIV` (Lean)). *Given an HQIV gauge potential A , a triality-indexed current family J , an octonion scalar Φ , a per-direction scalar slot Φ_{dir} , and a flavour bilinear assignment B , the record `SM_Lagrangian.fromHQIV` $A J \Phi \Phi_{\text{dir}} B$ sets*

$$\mathcal{L}_{\text{gauge}} = L_O^{\text{kin}}(A), \quad \mathcal{L}_{\text{fermion}} = \sum_{\text{gen}} L_O^{\text{src}}(J(\text{gen}), A),$$

$$\mathcal{L}_{\text{Higgs}}^{\text{kin}} = \sum_{\mu} \text{scalarNormSq}(\Phi_{\text{dir}}(\mu)), \quad \mathcal{L}_{\text{Higgs}}^{\text{pot}} = \text{higgsPotential}(\lambda_{\text{eff}}, \text{lockinVev}, \Phi),$$

$$\mathcal{L}_Y = \sum_{\ell \in \text{all_SM_flavours}} \mathcal{L}_{\text{Yukawa}}^{\text{SM}}(\ell, B(\ell)).$$

Theorem 6.3 (Headline: SM Lagrangian = HQIV discrete action). *For every $(A, J, \Phi, \Phi_{\text{dir}}, B)$, writing $L := \text{SM_Lagrangian_fromHQIV } A \ J \ \Phi \ \Phi_{\text{dir}} \ B$,*

- (1) $\mathcal{L}_{\text{gauge}}(L) = L_O^{\text{kin}}(A)$;
- (2) $\mathcal{L}_{\text{fermion}}(L) = L_O^{\text{src}}(J_{\text{SM}}^{\text{tot}}(J), A)$, where $J_{\text{SM}}^{\text{tot}}(J)(a, \nu) := \sum_{\text{gen}} J(\text{gen})(a, \nu)$ (J_SM_total);
- (3) $\mathcal{L}_{\text{Higgs}}^{\text{kin}}(L) = \sum_{\mu} \text{scalarNormSq}(\Phi_{\text{dir}}(\mu))$;
- (4) $\mathcal{L}_{\text{Higgs}}^{\text{pot}}(L) = \text{higgsPotential}(\lambda_{\text{eff}}, \text{lockinVev}, \Phi)$;
- (5) $\mathcal{L}_Y(L) = - \sum_{\ell \in \text{all_SM_flavours}} (\sqrt{2} \text{smMassFromGeometryLabel}(\ell) / \text{lockinVev}) B(\ell)$.

This is SM_Lagrangian_from_HQIV_discrete_action.

Theorem 6.4 (Total HQIV action split, Lean). *With $J_{\text{SM}}^{\text{tot}}(J)$ as in Theorem 6.3,*

$$\begin{aligned} & \text{action_total_general}(J_{\text{SM}}^{\text{tot}}(J), A, \phi_{\text{val}}, \rho_m, \rho_r) \\ &= \left(L_O^{\text{kin}}(A) + 4\pi \sum_{\text{gen}} L_O^{\text{src}}(J(\text{gen}), A) + L_O^{\phi}(A, \phi_{\text{val}}) \right) + S_{\text{HQVM grav}}(\phi_{\text{val}}, \rho_m, \rho_r). \end{aligned}$$

This is HQIV_total_action_eq_SM_gauge_fermion_plus_phi_coupling_plus_grav; its proof uses two invocations of L_O_source_general_add_J to split the SM total current into the three generation pieces, then a single ring on the residue.

Reading. The proved equality of Theorem 6.4 exposes the precise dictionary

$$S_{\text{HQIV}} = \mathcal{L}_{\text{gauge}}^{\text{SM}} + \frac{1}{4\pi} \mathcal{L}_{\text{fermion}}^{\text{SM}} + \mathcal{L}_{\text{Higgs cross}}^{\text{SM}} + S_{\text{HQVM grav}}$$

between the page-long SM Lagrangian and the HQIV total cell. The $\frac{1}{4\pi}$ factor is the discrete normalization carried over from the Maxwell-source convention $4\pi J \cdot A$ of `L_0_Maxwell_general`; the “ $\mathcal{L}_{\text{Higgs cross}}$ ” label captures the L_O^{ϕ} remnant of the textbook $(D_{\mu}\Phi)^{\dagger}(D^{\mu}\Phi)$ cross term once Φ is collapsed to the lattice scalar ϕ_{val} at the electroweak shell (`electroweakPhiShell`, `electroweakPhiShell_val=5` in `SM_GR_Unification.lean`).

7. PARAMETER ACCOUNTING AND WITNESSES

The full `SM_Lagrangian_fromHQIV` construction consumes the following *derived* inputs.

Theorem 7.1 (Parameter witness, Lean). $\alpha = 3/5$, $\gamma_{\text{HQIV}} = 2/5$, $\alpha_{\text{GUT}} = 1/42$ *simultaneously. This is SM_Lagrangian_parameter_witness, discharged by the triple $\langle \text{alpha_eq_3_5}, \text{gamma_eq_2_5}, \text{alpha_GUT_eq_1_42} \rangle$.*

Theorem 7.2 (Yukawa resonance witness, Lean). *For every SM flavour ℓ , when $\text{lockinVev} \neq 0$,*

$$\frac{y_{\text{SM}}(\ell) \text{lockinVev}}{\sqrt{2}} = m_{\tau}^{\text{Pl}} / \text{resonanceProduct}(\text{smGenerationIndex } \ell).$$

This is SM_Lagrangian_yukawa_resonance_witness.

TABLE 1. Parameter inputs to the HQIV-derived SM Lagrangian. None is a free fit; every value is a Lean-proved consequence of the two HQIV axioms (discrete null-cone counting and informational monogamy) plus the lock-in calibration.

Constant	HQIV source	Lean identifier
$\alpha = 3/5$	Lattice stars-and-bars normalization	<code>alpha_eq_3_5</code>
$\gamma_{\text{HQIV}} = 2/5$	Horizon split $\gamma = 1 - \alpha$	<code>gamma_eq_2_5</code>
$\alpha_{\text{GUT}} = 1/42$	$1/(\text{cube axes octonion imaginaries}) = 1/(6 \cdot 7)$	<code>alpha_GUT_eq_1_42</code>
λ_{eff}	Effective Higgs quartic at EW shell	<code>lambda_eff</code>
$v = \text{lockinVev}$	$\sqrt{\eta_{\text{paper}} \Omega_k(m_{\text{lockin}}; m_{\text{lockin}})}$	<code>lockinVev</code>
m_{τ}^{Pl}	τ mass in Planck units (lock-in lepton)	<code>m_tau_Pl</code>
$k_{\tau\mu}$	Charged-lepton resonance step $\tau \rightarrow \mu$	<code>resonance_k_tau_mu</code>
$k_{\mu e}$	Charged-lepton resonance step $\mu \rightarrow e$	<code>resonance_k_mu_e</code>

Remark 7.3 (Comparison with textbook parameter count). *The textbook SM has 19 free real parameters: 3 gauge couplings (g_1, g_2, g_3), 9 charged-fermion masses, 4 CKM angles, 2 Higgs parameters (μ^2, λ), 1 strong CP angle. With Dirac neutrinos add 3 masses and a 3×3 PMNS block. The HQIV bridge replaces: $g_1, g_2, g_3 \rightarrow$ same $\alpha_{\text{GUT}} + \phi$ -correction (`one_over_alpha_eff` of [1] / SM-GR module); charged-fermion masses $\rightarrow m_{\tau}^{\text{Pl}} \cdot k_{\tau\mu}^a k_{\mu e}^b$ ladder; CKM / PMNS $\rightarrow$ triality-orbit branching of `SMEembedding.lean`; $\mu^2, \lambda \rightarrow (\eta_{\text{paper}}, \lambda_{\text{eff}})$; strong CP $\rightarrow$ the algebra is real (no θ knob). The compressed count is three lattice rationals + one lepton-anchor scale + two resonance steps + one effective quartic; no fits.*

8. OUTLOOK: WHAT COMES NEXT

The structural bridge of Theorem 6.3 is exact at the bookkeeping level. To translate it to the textbook smooth-manifold field-equation form (a calculation-approximation layer for comparison with [5, 6], *not* a foundational completion of HQIV), three Tier-III tasks remain (ordering consistent with the roadmap of [1], Sect. §7):

- (1) **Variational fermion sector.** Promote the symbolic current $J : \text{Fin } 8 \rightarrow \text{Fin } 4 \rightarrow \mathbb{R}$ to a Dirac bilinear $\bar{\psi} \gamma^\mu \psi$ in a typed `Hqiv/Algebra/CliffordCl06Six...` ambient (the `Cl(0,6)` spinor carrier already lives in `Hqiv/Algebra/CliffordCl06SixStandardSpinorRho.lean`). The Euler–Lagrange identity for ψ then reads off the discrete Dirac equation; the abelian source coupling of Section 3 becomes the textbook minimal coupling.
- (2) **Non-abelian gauge kinetic for Weak and Strong.** Replace the abelian $F^a_{\mu\nu} = A^a_\nu - A^a_\mu$ in the Weak (Strong) sector by the non-abelian `weakF` (and a colour analogue), using the $\text{SU}(2)_L$ generators of `SMEembedding.lean` and the $\text{SU}(3)_c$ chart Lie law of `Hqiv/Physics/StrongColorSu3LieChartLaw.lean`. The reduction theorem `action_0_weak_higgs_reduces_exactly_to_action_0_Maxwell` already guarantees that the upgrade collapses to the proved abelian core when the commutator slot is zero.
- (3) **Higgs covariant derivative.** Wire `coordsGradientComponents` (from `Hqiv/Physics/ContinuumOmaxwellClosure.lean`) into both the fermion bilinear and the Higgs scalar so

that D_μ contains the full (A_μ, W_μ, B_μ) data; verify that the resulting kinetic shadow reduces to the diagonal `scalarNormSq` sum in the abelian limit.

None of these tasks requires a new physics input, and none is an HQIV “open problem”: the constants, vev, and mass ladder are already proved on the discrete spine, and the smooth-manifold form is purely a translation layer for comparison with continuum-QFT textbooks. The remaining work is the algebraic upgrade of the symbolic “slot” fields to typed bundles inside the existing Lean infrastructure — not a continuum-completion programme.

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